Shortest-Path Competition Solver Report

Final solver: CSR + exact ALT landmarks + bucket queues

Interface and Correctness

- Required CLI kept: ./solver <graph_file> <query_file> <output_file>.
- The implementation is derived from foundation Dijkstra and returns exact shortest-path distances.
- The final full local grade.py run reported ok on all 12 released instances.

Main Optimizations

- CSR adjacency arrays, whole-file input, buffered output, and generation counters reduce overhead.
- Exact ALT/A* uses 24 farthest-first landmarks, including SOCIAL graphs without coordinates.
- GRID/SOCIAL low-diameter graphs use Dial buckets and compact int16 landmark tables when safe.
- ROAD uses unidirectional ALT plus bitmap buckets; CLUSTER uses bidirectional ALT with an 8-ary heap.
- SOCIAL initializes a landmark-derived upper bound min_k d(s,L_k)+d(L_k,t) for exact-safe pruning.

Full Local Grade.py Measurements

Command: python3 grade.py --foundation ./foundation --solver ./solver --instances instances.txt --progress

Cat	Instance	T_base	T_solver	Speedup	Note
ROAD	road_small.graph	4.3011	0.1951	22.042	ok
GRID	grid_small.graph	4.4041	0.0952	46.256	ok
CLUSTER	cluster_small.graph	4.9962	0.3745	13.341	ok
SOCIAL	social_small.graph	13.2219	0.1994	66.296	ok
ROAD	road_medium.graph	62.7959	1.4695	42.733	ok
GRID	grid_medium.graph	47.1362	0.7804	60.401	ok
CLUSTER	cluster_medium.graph	76.3894	1.6086	47.487	ok
SOCIAL	social_medium.graph	244.1336	3.0551	79.910	ok
ROAD	road_large.graph	904.8340	13.7633	65.743	ok
GRID	grid_large.graph	587.3162	9.4342	62.254	ok
CLUSTER	cluster_large.graph	1079.0337	10.6161	100.000	ok
SOCIAL	social_large.graph	3620.3446	21.6503	100.000	ok

Geometric Means

CLUSTER 39.865 (n=3) GRID 55.821 (n=3) ROAD 39.563 (n=3) SOCIAL 80.915 (n=3)

Algorithm track 44.486 (n=9) Engineering track 80.915 (n=3) Overall 51.662 (n=12)

Comparison and Reproducibility

Compared with the fastest reference-style local large run before SOCIAL ALT:
road 14.0391->13.7633 s, grid 9.6468->9.4342 s, cluster 11.3167->10.6161 s, social 68.5482->21.6503 s.
Large-tier geometric improvement over that local reference is about 1.36x.
Prefetch/SIMD intrinsics were left out so the final code remains ISO C++17/std-library-only.